

Breast Cancer Classification

1. Overview

This project implements a binary classification model (Logistic Regression) to predict whether a tumor is Malignant (0) or Benign (1) based on features extracted from digitized images of breast mass.

2. Dataset Description

- **Source:** Scikit-learn's built-in `load_breast_cancer()` dataset.
- **Features (X):** 30 numeric attributes (e.g., radius, texture and perimeter).
- **Target (y):** Binary classification (0: Malignant, 1: Benign).
- **Data Split:** 70% Training and 30% Testing (Random State: 42).

3. Model Implementation:

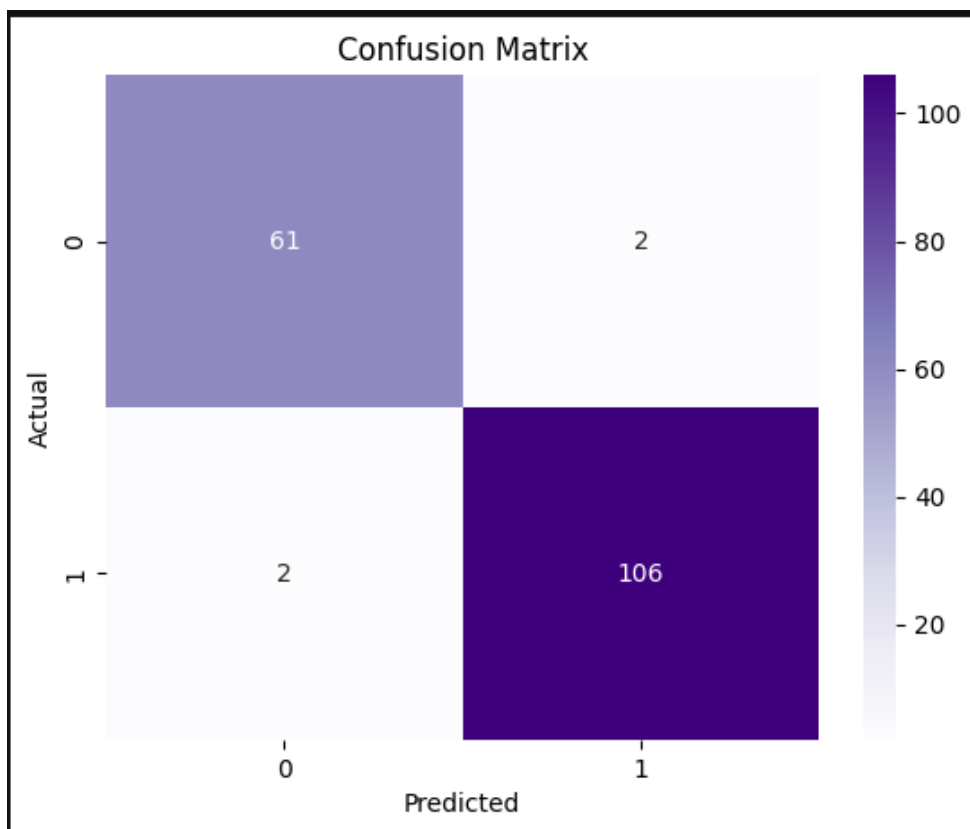
The model uses Logistic Regression, which is a powerful algorithm for binary classification tasks.

- **Max Iterations:** Set to 5000 to ensure the solver converges.
- **Training:** The model was fitted on the training set (`X_train`, `y_train`).

4. Performance Analysis:

- **Confusion Matrix:**

The confusion matrix shows how many predictions were correct versus incorrect:



- **Accuracy Score:**

The model achieved an accuracy of approximately **97.66%** on the test data.